

Question:

Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .

Reasoning + Answer:

<think>

1. We need to find all integer bases b greater than 9 such that the number 17_b (which is 17 in base b) is a divisor of 97_b (which is 97 in base b).

2. Convert the numbers 17_b and 97_b to base 10.

- 17_b in base 10 is $1 \cdot b + 7 = b + 7$.

- 97_b in base 10 is $9 \cdot b + 7 = 9b + 7$.

3. According to the problem, $b + 7$ should be a divisor of $9b + 7$. We can write this as:

$$\frac{9b + 7}{b + 7}$$

should be an integer.

4. Perform polynomial long division on $9b + 7$ by $b + 7$:

$$\begin{array}{l} - (9b \div b = 9) \text{ (first term of quotient)} \\ - (9b + 7 - (9b + 6) = 1) \end{array}$$

5. The quotient is 9 and the remainder is 1, giving:

$$9b + 7 = (b + 7) \cdot 9 + 1$$

6. For $b + 7$ to divide $9b + 7$ exactly, the remainder must be 0:

$$(b + 7) \cdot 9 + 1 \rightarrow 1$$

7. Simplifying gives:

$$9b + 7 \equiv 0 \pmod{b + 7}$$

8. This leads to:

$$9b + 7 = (b + 7) \cdot 9$$

Simplify:

$$9b + 7 = 9b + 63 \rightarrow 7 \div 7 = 1.$$

9. Therefore:

$$b + 7 \text{ must be a factor of } 1 \text{ i.e. } b + 7 = 1.$$

10. This implies $b + 7 = 1$ implies the integer b greater than 9.

11. Let's determine exact values then satisfies b and integer, no integers here satisfies.

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<answer>

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64,